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The Editor-in-Chief
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Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled *"EASFL: Energy-Aware, Poisoning-Resilient Federated Learning for Anomaly Detection and Lifetime Extension in Wireless Sensor Networks"* for consideration as a regular paper in the IEEE Transactions on Green Communications and Networking.

Federated learning is increasingly proposed for on-device anomaly detection in the Internet of Things, yet in battery-powered wireless sensor networks the radio energy spent exchanging model updates can exceed that of forwarding the raw data. Our paper treats energy as the primary design constraint of a federated protocol and measures its cost in joules and in network lifetime, which places it squarely within the journal's scope on energy-efficient networking and green machine learning for communication systems.

The main contributions are:

1. A cluster-based federated learning protocol in which cluster-head selection, participant selection, update compression (top-k sparsification, 8-bit quantization, error feedback) and aggregation are each chosen to reduce radio energy under an explicit bit-level first-order radio model.
2. A two-tier robust aggregation scheme with reputation-based quarantine that needs no trusted root dataset, together with a zero-communication on-node personalization step that closes the non-IID accuracy gap.
3. An energy-coupled evaluation, including a network-level F1-score in which depleted nodes raise no alarms, ablations, four poisoning attacks against three robust-aggregation baselines, sensitivity analyses, commercial-radio validation and a real-data evaluation on the Intel Berkeley Lab deployment. All code, seeds and raw results are publicly available at `https://github.com/vadlashiva/EASFL`.

In a 100-node network, EASFL reduces over-the-air traffic by 13× and energy use by 3× relative to hierarchical FedAvg, delays the first node death from round 35 to round 547, and attains a network-level F1-score of 0.82 at round 200, against 0.26 for hierarchical FedAvg and 0.70 for centralized learning. On real readings and measured links from the Intel Berkeley Lab deployment, EASFL matches the accuracy of centralized and federated baselines with 20× less traffic and keeps every mote alive at round 200, and with CC2420 and CC2650 radio parameters under IEEE 802.15.4 framing it uses 3.4–4.8× less energy per round than hierarchical FedAvg.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. Both authors, Dr. Shiva Kumar V and Dr. Choupiri Shivakeshi, have

approved the manuscript and agree with its submission. We have no conflicts of interest to declare.

Thank you for considering our work. We look forward to your response.

Sincerely,

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On behalf of co-author Dr. Choupiri Shivakeshi